

D5 Geometric sequences A level only

Name: _____

Class: _____

Date: _____

Time: **146 minutes**

Marks: **123 marks**

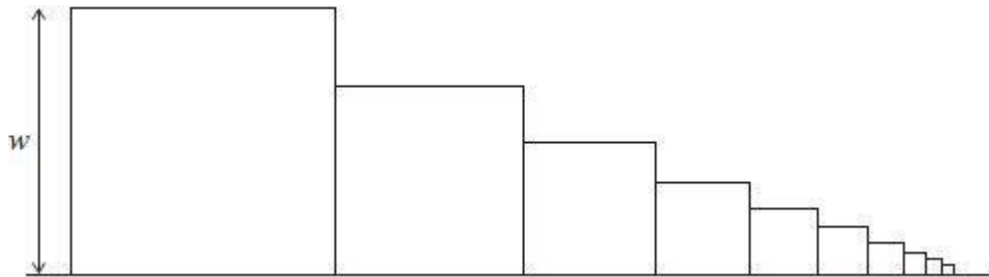
Comments:

EXAM PAPERS PRACTICE

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Q1.

Helen is creating a mosaic pattern by placing square tiles next to each other along a straight line.



The area of each tile is half the area of the previous tile, and the sides of the largest tile have length w centimetres.

- (a) Find, in terms of w , the length of the sides of the second largest tile.

(1)

- (b) Assume the tiles are in contact with adjacent tiles, but do not overlap.

Show that, no matter how many tiles are in the pattern, the total length of the series of tiles will be less than $3.5w$.

(4)

- (c) Helen decides the pattern will look better if she leaves a 3 millimetre gap between adjacent tiles.

Explain how you could refine the model used in part (b) to account for the 3 millimetre gap, and state how the total length of the series of tiles will be affected.

(2)

(Total 7 marks)

Q2.

- (a) Three consecutive terms in an arithmetic sequence are $3e^{-p}$, 5, $3e^p$

Find the possible values of p . Give your answers in an exact form.

(6)

- (b) Prove that there is no possible value of q for which $3e^{-q}$, 5, $3e^q$ are consecutive terms of a geometric sequence.

(4)

(Total 10 marks)

Q3.

- (a) A geometric series begins $420 + 294 + 205.8 + \dots$

- (i) Show that the common ratio of the series is 0.7.

(1)

(ii) Find the sum to infinity of the series.

(2)

(iii) Write the n th term of the series in the form $p \times q^n$, where p and q are constants.

(2)

(b) The first term of an arithmetic series is 240 and the common difference of the series is -8 .

The n th term of the series is u_n .

(i) Write down an expression for u_n .

(1)

(ii) Given that $u_k = 0$, find the value of $\sum_{n=1}^k u_n$.

(4)

(Total 10 marks)

Q4.

A geometric series has first term 80 and common ratio $\frac{1}{2}$.

(a) Find the third term of the series.

(1)

(b) Find the sum to infinity of the series.

(2)

(c) Find the sum of the first 12 terms of the series, giving your answer to two decimal places.

(2)

(Total 5 marks)

Q5.

The n th term of a geometric series is u_n , where $u_n = 48 \left(\frac{1}{4}\right)^n$.

(a) Find the value of u_1 and the value of u_2 .

(2)

(b) Find the value of the common ratio of the series.

(1)

(c) Find the sum to infinity of the series.

(2)

- (d) Find the value of $\sum_{n=4}^{\infty} u_n$.

(3)

(Total 8 marks)

Q6.

A geometric series has third term 36 and sixth term 972.

- (a) (i) Show that the common ratio of the series is 3.

(2)

- (ii) Find the first term of the series.

(2)

- (b) The n th term of the series is u_n .

- (i) Show that $\sum_{n=1}^{20} u_n = 2(3^{20} - 1)$.

(2)

- (ii) Find the least value of n such that $u_n > 4 \times 10^{15}$.

(3)

(Total 9 marks)

Q7.

The first term of a geometric series is 12 and the common ratio of the series is $\frac{3}{8}$.

- (a) Find the sum to infinity of the series.

(2)

- (b) Show that the sixth term of the series can be written in the form $\frac{3^8}{2^{13}}$.

(3)

- (c) The n th term of the series is u_n .

- (i) Write down an expression for u_n in terms of n .

(1)

- (ii) Hence show that

$$\log_a u_n = n \log_a 3 - (3n - 5) \log_a 2$$

(4)

(Total 10 marks)

Q8.

- (a) An infinite geometric series has common ratio r .

The first term of the series is 10 and its sum to infinity is 50.

(i) Show that $r = \frac{4}{5}$.

(2)

- (ii) Find the second term of the series.

(2)

- (b) The first and second terms of the geometric series in part (a) have the same values as the 4th and 8th terms respectively of an arithmetic series.

- (i) Find the common difference of the arithmetic series.

(3)

(ii) The n th term of the arithmetic series is u_n . Find the value of $\sum_{n=1}^{40} u_n$.

(4)

(Total 11 marks)

Q9.

A geometric series has second term 375 and fifth term 81.

- (a) (i) Show that the common ratio of the series is 0.6.

(3)

- (ii) Find the first term of the series.

(2)

- (b) Find the sum to infinity of the series.

(2)

(c) The n th term of the series is u_n . Find the value of $\sum_{n=6}^{\infty} u_n$.

(4)

(Total 11 marks)

Q10.

A geometric series begins

$$20 + 16 + 12.8 + 10.24 + \dots$$

- (a) Find the common ratio of the series.

(1)

- (b) Find the sum to infinity of the series. (2)
- (c) Find the sum of the first 20 terms of the series, giving your answer to three decimal places. (2)
- (d) Prove that the n th term of the series is 25×0.8^n . (2)
- (Total 7 marks)

Q11.

The second term of a geometric series is 48 and the fourth term is 3.

- (a) Show that one possible value for the common ratio, r , of the series is $-\frac{1}{4}$ and state the other value. (4)
- (b) In the case when $r = -\frac{1}{4}$, find:
- (i) the first term; (1)
- (ii) the sum to infinity of the series. (2)
- (Total 7 marks)

Q12.

The n th term of a geometric sequence is u_n , where

$$u_n = 3 \times 4^n$$

- (a) Find the value of u_1 and show that $u_2 = 48$. (2)
- (b) Write down the common ratio of the geometric sequence. (1)
- (c) (i) Show that the sum of the first 12 terms of the geometric sequence is $4^k - 4$, where k is an integer. (3)
- (ii) Hence find the value of $\sum_{n=2}^{12} u_n$. (1)

(Total 7 marks)

Q13.

- (a) Use logarithms to solve the equation $0.8^x = 0.05$, giving your answer to three decimal places.

(3)

- (b) An infinite geometric series has common ratio r . The sum to infinity of the series is five times the first term of the series.

- (i) Show that $r = 0.8$.

(3)

- (ii) Given that the first term of the series is 20, find the least value of n such that the n th term of the series is less than 1.

(3)

(Total 9 marks)

Q14.

The sum to infinity of a geometric series is four times the first term of the series.

- (a) Show that the common ratio, r , of the geometric series is $\frac{3}{4}$.

(3)

- (b) The first term of the geometric series is 48. Find the sum of the first 10 terms of the series, giving your answer to four decimal places.

(2)

- (c) The n th term of the geometric series is u_n and the $(2n)$ th term of the series is u_{2n} .

- (i) Write u_n and u_{2n} in terms of n .

(2)

- (ii) Hence show that $\log_{10}(u_n) - \log_{10}(u_{2n}) = n \log_{10}\left(\frac{4}{3}\right)$.

(3)

- (iii) Hence show that the value of $\log_{10}\left(\frac{24_{100}}{24_{200}}\right)$

is 12.5 correct to three significant figures.

(2)

(Total 12 marks)